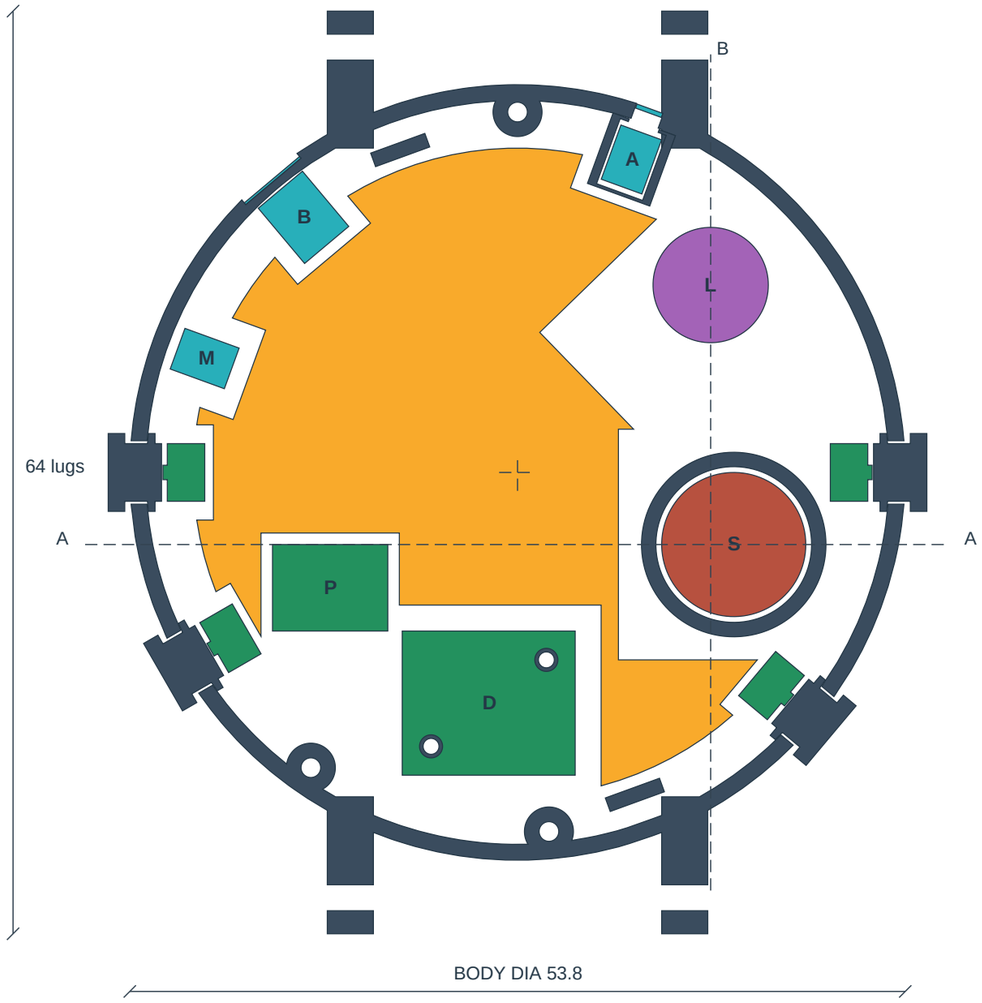
UNHYGIENIX / LAYOUT COMPARISON

CONCEPT 01 | UNVERIFIED COMPONENT FIT | BATTERY UNSELECTED

A / minimum-thickness study

D53.8 x H13.9; assumes bare host H6.0

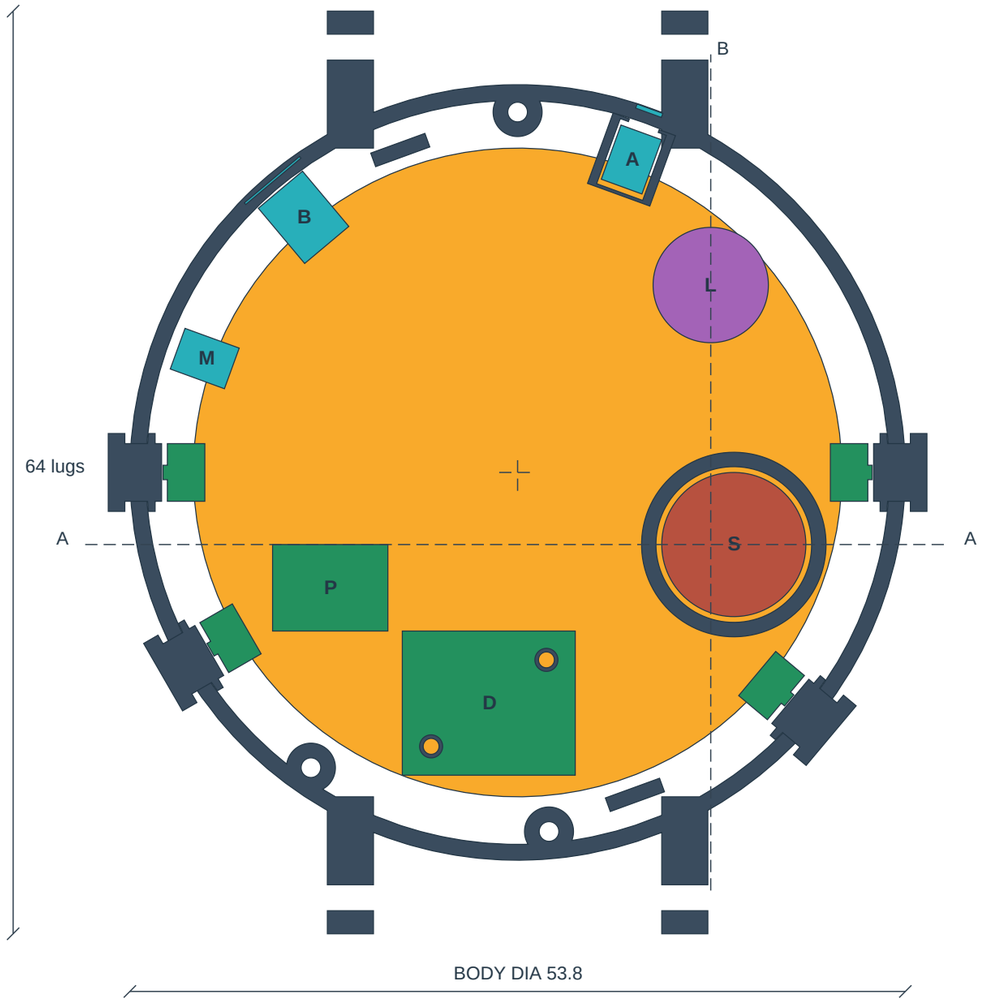


Cell outline area 761 mm2 (47.8% of D45)

Host hidden; footprints at different Z overlaid.

B / full-cell stacked concept

D53.8 x H19.5; assumes bare host H6.0



Cell outline area 1590 mm2 (100.0% of D45)

Host hidden; footprints at different Z overlaid.

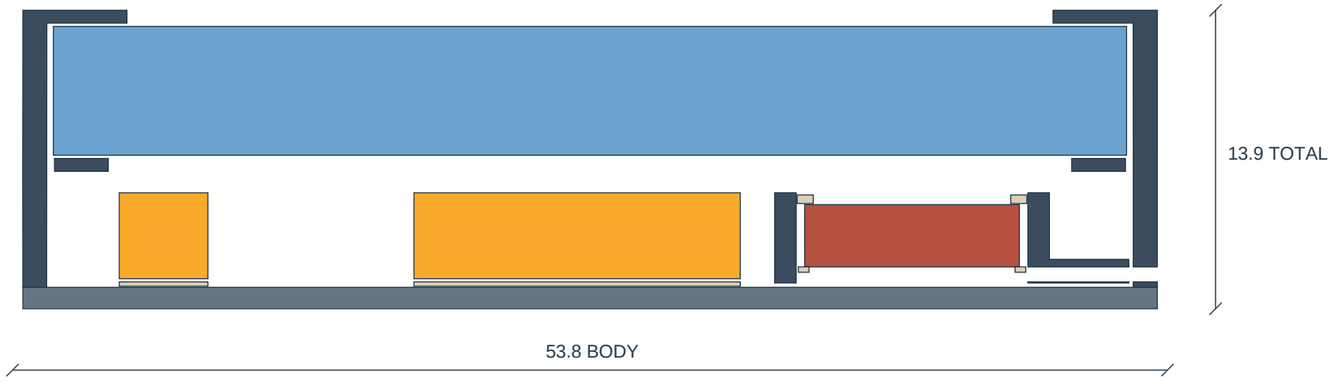
S speaker L LRA D daughterboard P pack protection M compass B barometer A ambient light

A has no demonstrated 750-800 mAh pack. B preserves cell area at a 5.6 mm thickness penalty; pack remains unselected.

UNHYGIENIX / A / NOTCHED CELL SECTIONS

CONCEPT 01 | UNVERIFIED COMPONENT FIT | BATTERY UNSELECTED

A-A / y = -5.0, looking toward +Y / speaker face and side duct



Height equation: H = 7.9 + measured host height

Host solid is a D51 x H6.0 reservation, not a bare-PCB model.

Host lower datum Z 7.00; solid starts Z 7.15.

0.15 axial gap each side; 0.30 nominal radial host clearance.

0.60 bezel; D44.30 aperture over published 43.76 active area.

Cell: Z1.40-5.40, 4.00 target; no selected part/tolerance.

Installed cell reservation: Z1.20-6.70, including growth/liners.

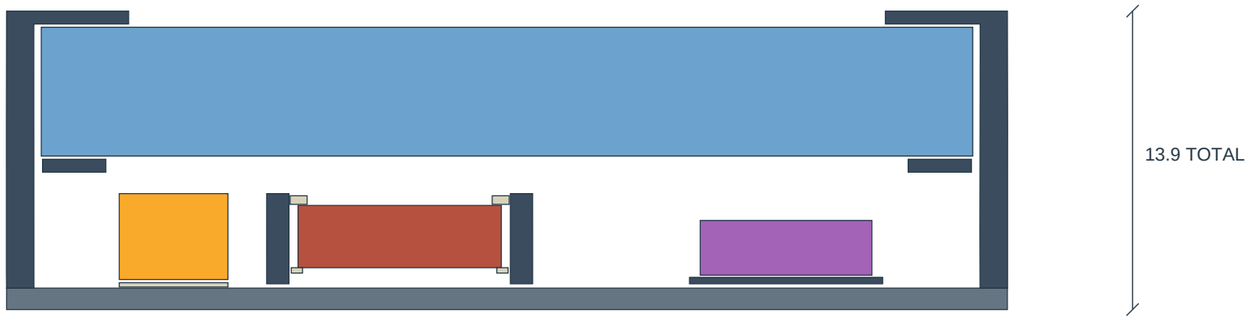
Speaker body Z1.95-4.85; reserved H2.90.

Cup top Z5.40; gasket H0.25; face plenum H0.50.

Outlet duct 2.4 x 0.7; cavity and grille not acoustically qualified.

Rear floor 1.00; dielectric liner H0.20; no pouch support loads.

B-B / x = 13.4, looking toward -X / LRA and battery zone



LRA body Z1.60-4.15; D8.0 x H2.55 nominal.

Full FPC reservation 9.2 x 14.2, tail directed inward.

Rigid 0.30 shoe bonds to rear/deck, separately from cell.

0.10 adhesive under LRA body is an assumed stack allowance.

Optional rear HR: external ghost only, zero internal allocation.

No optical window or HR sensor is silently fitted in the cell volume.

A: concave cell outline is a feasibility envelope only.

Supplier manufacturability/capacity is unknown.

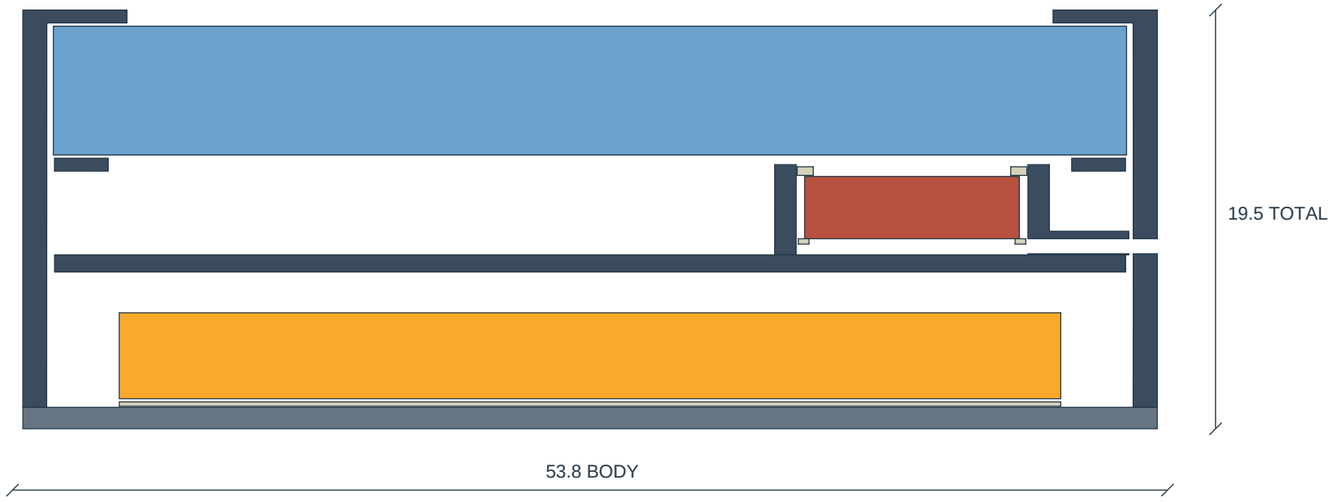
No rectangular 48 x 46 mm pocket is assumed.

Section boundaries tessellated from CAD at 0.04 mm; dimensions are exact model parameters. Colors identify placeholder components.

UNHYGIENIX / B / FULL CELL SECTIONS

CONCEPT 01 | UNVERIFIED COMPONENT FIT | BATTERY UNSELECTED

A-A / y = -5.0, looking toward +Y / speaker face and side duct



Height equation: H = 13.5 + measured host height

Host solid is a D51 x H6.0 reservation, not a bare-PCB model.

Host lower datum Z 12.60; solid starts Z 12.75.

0.15 axial gap each side; 0.30 nominal radial host clearance.

0.60 bezel; D44.30 aperture over published 43.76 active area.

Cell: Z1.40-5.40, 4.00 target; no selected part/tolerance.

Installed cell reservation: Z1.20-6.70, including growth/liners.

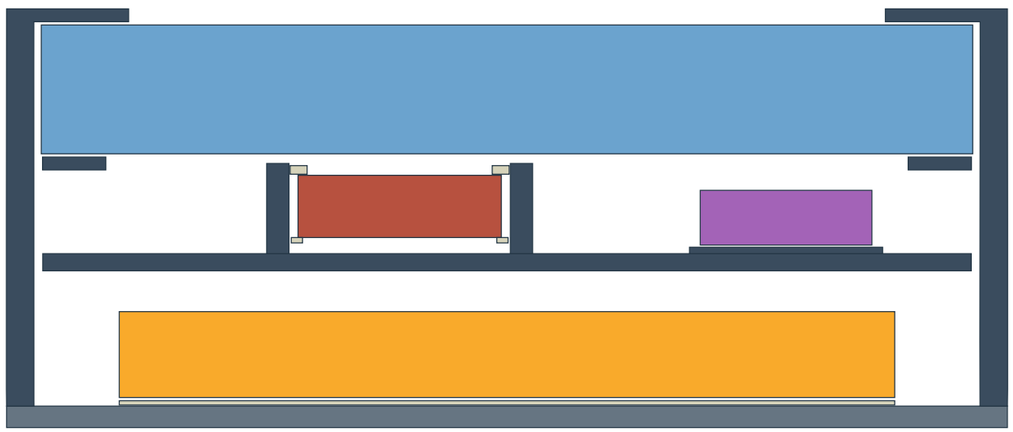
Speaker body Z8.85-11.75; reserved H2.90.

Cup top Z12.30; gasket H0.25; face plenum H0.50.

Outlet duct 2.4 x 0.7; cavity and grille not acoustically qualified.

Rear floor 1.00; dielectric liner H0.20; no pouch support loads.

B-B / x = 13.4, looking toward -X / LRA and battery zone



LRA body Z8.50-11.05; D8.0 x H2.55 nominal.

Full FPC reservation 9.2 x 14.2, tail directed inward.

Rigid 0.30 shoe bonds to rear/deck, separately from cell.

0.10 adhesive under LRA body is an assumed stack allowance.

Optional rear HR: external ghost only, zero internal allocation.

No optical window or HR sensor is silently fitted in the cell volume.

B: deck Z7.30-8.10; supported on case boss collars.

Remote protection bay Z8.70-11.70 above deck.

Pack lead length and construction require supplier approval.

Section boundaries tessellated from CAD at 0.04 mm; dimensions are exact model parameters. Colors identify placeholder components.

UNHYGIENIX / BUTTON / PCB / DATUM DETAILS

CONCEPT 01 | UNVERIFIED COMPONENT FIT | BATTERY UNSELECTED

C / radial button section at angle 0 deg, A layout / scale 12:1



Actual section of housing, captured key and switch reservation.

Rod 4.00 x 1.40 in 4.40 x 1.80 guide: 0.20 running gap per side.

Inside flange 5.40 x 2.40 retained by aperture; radius trimmed to clear wall.

Outside shoulder limits inward travel to 0.35 nominal.

Rest rod-to-switch-tip gap 0.10. Assumed electrical travel 0.15:

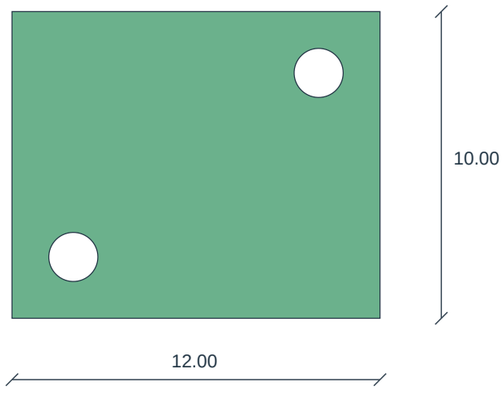
0.25 total to actuate, leaving 0.10 to hard stop. Actual switch TBD.

Housing supplies retention/stop, selected switch must supply return force.

Back/Select positions are proposals; engagement with stock switches unverified.

Switch carrier, transfer geometry, seals and PCB routing are not released.

D / proposed daughterboard outline / scale 5:1



Global board extents X[-8,4], Y[-21,-11].

Holes D1.60 at (-6,-19), (2,-13). Z1.80 A / Z8.70 B.

Case datums and development access

O = display center; Z0 = external rear surface; +Y = 12 o'clock lug.

Three rear fasteners on R25 at 90, 235, 275 deg.

Pilot D1.36 x 4.5; rear clearance D1.90; head recess D3.30 x 0.55.

Candidate small polymer-thread screws; length/pullout not qualified.

USB at 270 deg: 14 x 4.6 case opening, 23 x 14 x 7 plug keepout.

Reset-tool opening at 250 deg, 2 x 2. Actual USB/reset positions TBD.

Lugs: 20.0 clear width, 59.0 bar spacing, 64.0 tip span, D1.80 bores.

Strap sizes, fit, pull strength and wrist clearance are prototype assumptions.

Printed PA12 nominal +/-0.20 on dimensions; calibrate a coupon first.

Dry bench prototype only. No water-resistance or final fastening claim.

UNHYGIENIX / CLEARANCE / EVIDENCE / RELEASE GATES

CONCEPT 01 | UNVERIFIED COMPONENT FIT | BATTERY UNSELECTED

What is established

Manufacturer image: D51 assembly, 43.76 visible width, stock side 11.70 / 12.10.

Downloaded host resource contains DWG + PDF, no STEP solid. PDF label is ESP32-S3.

It cannot establish actual T5 PCB revision, internal heights or mounting datums.

LRA drawing: D8.0 +/-0.1, H2.55 +/-0.05; FPC overall 13.265 +/-0.3 x 8.535 +/-0.3.

Soberton product and 2019 drawing show D10 x H2.4; this model retains H2.9 reserve.

Candidate dimensions are not installed-fit verification or final component selection.

Numerical CAD checks

thin_notched: 36 physical parts; 0 positive-volume intersections; STEP reimport valid = True.

13 printable meshes; watertight = True. Minimum installed-cell gap 0.20 to PRINT_rear_cover.

full_cell_stacked: 37 physical parts; 0 positive-volume intersections; STEP reimport valid = True.

14 printable meshes; watertight = True. Minimum installed-cell gap 0.20 to PRINT_rear_cover.

Open release gates - a clear placeholder assembly is not proof of component fit

FPC, RF and actual PCB geometry remain unregistered. Red keepout STEP and JSON record their intersections.

Register actual flex before routing: current illustrative FPC intersects battery/retainer and optional pressure island.

Illustrative RF sector intersects LRA. Obtain antenna drawing and rearrange magnetic/metal parts before fit release.

Battery pack, lead exit, growth, protection and capacity must be supplier-drawn; A removes about half the cell area.

Magnetic center distances are geometric only; test speaker/LRA drive, charging, screws and compass saturation.

Qualify button return/travel, carrier, USB plug and recovery access, vent adhesive, speaker response and strap loads.

HR requires a separately budgeted rear cover/stack. Charging current and enclosure temperature need fitted-cell review.

All STEP/STL outputs are concept fit-study artifacts. No parts fabricated, ordered or fitted; firmware untouched by this design.